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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 06/26/2025 |
| **PI Name:** | Tabassum, Shawana Ph.D. | | |
| **Project Title:** | Leaf Sensor Network for In Situ and Multiparametric Analysis of Crop Stressors | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 75799 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies |
| **Start Date:** | 03/01/2023 | **End Date:** | 02/28/2025 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | O'Rourke, Aubrie | **Contact Phone:** | |
| **Contact Email:** | aubrie.e.orourke@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 02/28/2025 per NSSC information (Ed., 12/30/2024). | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Rajan, Nithya Ph.D. ( Texas A&M University ) | | |
| **Grant/Contract No.:** | 80NSSC23K0401 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | This project aims to advance the fundamental understanding of the hormonal responses in plants in a spaceflight-like environment through an in situ technology that collects and analyzes data on plant phytohormones in real time. Progressive variations in the phytohormone levels initiate a plant's defense mechanisms against environmental stressors. Salicylic acid, jasmonic acid, abscisic acid, and indole-3-acetic acid are among the most important regulators of induced defense mechanisms. Progressive variations in their levels have been reported under many drought and cold/heat-stressed conditions on Earth. However, the dynamic interaction mechanism of these hormones is not fully elucidated in a "space farming" setting due to the lack of technology, needed to facilitate in situ sensing. Real-time understanding of a plant's responses to stressors is essential to minimize stress-induced growth and yield declines in plants. Toward this end, this project develops a lightweight and integrated stem sensor with sensing elements to monitor plant hormonal variations in real-time. The system comprises a multiplexed hormone sensor for quantitatively measuring the primary defense hormones: Salicylic acid and abscisic acid. The impact of the following stressors on the hormone levels is analyzed: changes in temperature, lighting, and growth media. In contrast to the traditional discrete, disruptive, in vitro, time-intensive, and heavyweight instruments used for molecular analysis, our proposed sensor is plant-wearable and provides in situ monitoring capabilities. We have developed functional correlations of the measured hormonal variations with physiological indicators (photosynthesis, respiration, and transpiration) to differentiate the effect of growing conditions on individual plant responses. The knowledge gained from this project will advance future research on predicting and improving plant growth and productivity under spaceflight stressors. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This study aims to enhance our understanding of a plant's defense mechanisms against a combination of environmental stressors by providing real-time data on variations in the plant's defense hormones. Accurate and real-time measurements of plant parameters are expected to provide agricultural producers with immediate insights into crop stress levels, facilitating precise adjustments in irrigation and fertilizer application. The outcomes of this project are poised to have a lasting positive impact on terrestrial agriculture by: (1) providing a tool for early assessment of crop water and nutrient uptake, (2) contributing to the existing and new databases on irrigation and fertilizer management, modeling data, and linking local data with national-level databases, (3) developing science-based irrigation and fertilizer management policies, and (4) improving the environmental sustainability of agricultural practices. |
| **Task Progress:** | This project aimed to advance the understanding of how plants respond to environmental stress in spaceflight-like conditions by developing a novel sensor platform capable of monitoring hormone levels in real time. Hormones such as salicylic acid (SA) and abscisic acid (ABA) play key roles in plant responses to stress, including drought, heat, and limited nutrients. However, existing plant monitoring tools rely on periodic sampling or visual observation, which do not capture dynamic hormone changes over time. Our work bridges this gap by introducing a non-destructive, in-situ sensor system that allows for continuous tracking of key hormonal signals inside the plant stem.<br>During the first year, the team successfully designed and fabricated a 3D-printed electrochemical sensor to detect salicylic acid in the sap of cowpea plants, a model legume crop. The sensors were validated in the lab using known concentrations of salicylic acid and demonstrated consistent and repeatable performance across multiple sensors.<br><br>To improve specificity, the team tested the sensor's ability to distinguish SA from other common plant hormones, including abscisic acid, indole-3-acetic acid, and gibberellic acid. Results showed that the sensor responded strongly to SA while showing minimal interference from other hormones, confirming its selectivity.<br><br>In the second year, the sensor platform was expanded to include detection of abscisic acid (ABA), another stress-related hormone. Additionally, the overall sensor size was reduced by 50%, resulting in minimal damage to the plants during data collection.<br><br>Parallel to sensor development, experiments were conducted in controlled growth chambers and greenhouse settings at Texas A&M University. Cowpea plants were grown under various soil and hydroponic media, simulating stress conditions related to water and nutrient availability. Physiological traits such as photosynthesis rate, transpiration, and stomatal conductance were measured to assess the plants' responses to these environments.<br><br>The newly developed sensors were then tested directly in the stems of these plants. Results showed that the sensors successfully measured variations in SA and ABA levels under different growing conditions. Plants in peat-perlite media showed the highest hormone concentrations—indicating water stress—while those in hydroponics had the lowest, consistent with better water availability. These hormonal trends were in agreement with the physiological measurements and plant biomass outcomes, further validating the sensor system.<br><br>Additionally, calibration plots were re-established inside the growth chamber to account for environmental influences on sensor response. Preliminary correlation analysis also showed a relationship between hormone levels and physiological performance, laying the groundwork for future predictive models.<br><br>The project's main innovation—real-time plant hormone sensing—marks a significant step forward in space agriculture. Unlike conventional methods that rely on lab-based sampling, this integrated sensor system provides immediate insights into plant health and stress. This capability is especially valuable for closed environments like spacecraft or extraterrestrial greenhouses, where early detection of stress could be critical for crop productivity and mission success.<br><br>In summary, the project achieved all major milestones: a robust, miniaturized hormone sensor was developed and validated; field-like testing environments confirmed the sensor's performance; and meaningful physiological-hormonal correlations were established. The work offers a foundation for next-generation plant monitoring systems designed to support sustainable agriculture in space and other extreme environments. |
| **Bibliography Type:** | Description: (Last Updated: 08/06/2025) |
| **Abstracts for Journals and Proceedings** | Pali M, Islam MN, Poudyal C, Wheeler W, Rajan N, Tabassum S. "SMART-PH: Sensing Modality to Assist Real-time Tracking of Plant Health in spaceflight-like conditions." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024.<br>Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024 |

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| **Abstracts for Journals and Proceedings** | Poudyal C, Wheeler W, Pali M, Islam MN, Tabassum S, Rajan N. "Exploring the phytohormonal responses of plants for space agriculture using in-situ sensor technology." CANVAS (2024 ASA, CSSA, SSSA International Annual Meeting), San Antonio, Texas, November 10-13, 2024.<br>Abstracts. CANVAS (2024 ASA, CSSA, SSSA International Annual Meeting), San Antonio, Texas, November 10-13, 2024. , Nov-2024 |
| **Abstracts for Journals and Proceedings** | Poudyal C, Wheeler W, Pali M, Islam MN, Tabassum S, Rajan N. "Real-time monitoring of plant stress hormones for extra-terrestrial agriculture." 2024 Data Science Space Week: From Earth to Mars - Observations in Biological and Physical Sciences, College Station, Texas, April 1-5, 2024.<br>Abstracts. 2024 Data Science Space Week: From Earth to Mars - Observations in Biological and Physical Sciences, College Station, Texas, April 1-5, 2024. , Apr-2024 |